

| D101   BENZENE SEPARATION |             |                  |             |         |
|---------------------------|-------------|------------------|-------------|---------|
| Temp C                    | Distil Flow | Benz Composition | Benz Amount | Unit    |
| 118                       | 22.42       | 0.9915           | 22.22943    | kmol/hr |
| 119                       | 38.41       | 0.9917           | 38.091197   | kmol/hr |
| 120                       | 53.58       | 0.9919           | 53.146002   | kmol/hr |
| 121                       | 67.96       | 0.9921           | 67.423116   | kmol/hr |
| 122                       | 81.67       | 0.9922           | 81.032974   | kmol/hr |
| 123                       | 94.7        | 0.9923           | 93.97081    | kmol/hr |
| 124                       | 107.1       | 0.9922           | 106.26462   | kmol/hr |
| 125                       | 119         | 0.9922           | 118.0718    | kmol/hr |
| 126                       | 130.3       | 0.992            | 129.2576    | kmol/hr |
| 127                       | 141.3       | 0.9916           | 140.11308   | kmol/hr |
| 128                       | 151.6       | 0.9911           | 150.25076   | kmol/hr |
| 130                       | 171.7       | 0.9883           | 169.69111   | kmol/hr |
| 132                       | 192         | 0.9732           | 186.8544    | kmol/hr |
| 135                       | 255.8       | 0.7735           | 197.8613    | kmol/hr |

| Varying R from 1.5 to 2.5   T = 126 C |             |           |             |         |
|---------------------------------------|-------------|-----------|-------------|---------|
| R                                     | Distil Flow | Benz Comp | Amount Benz | Unit    |
| 1.5                                   | 132.8       | 0.97      | 128.816     | kmol/hr |
| 1.6                                   | 132.5       | 0.9748    | 129.161     | kmol/hr |
| 1.7                                   | 131.9       | 0.9789    | 129.11691   | kmol/hr |
| 1.8                                   | 131.5       | 0.982     | 129.133     | kmol/hr |
| 1.9                                   | 131.2       | 0.9844    | 129.15328   | kmol/hr |
| 2                                     | 131.2       | 0.9863    | 129.40256   | kmol/hr |
| 2.1                                   | 130.8       | 0.9879    | 129.21732   | kmol/hr |
| 2.2                                   | 130.7       | 0.9892    | 129.28844   | kmol/hr |
| 2.3                                   | 130.6       | 0.9903    | 129.33318   | kmol/hr |
| 2.4                                   | 130.4       | 0.9912    | 129.25248   | kmol/hr |
| 2.5                                   | 130.4       | 0.992     | 129.3568    | kmol/hr |

| D102   TOLUENE SEPARATION |          |                 |            |         |
|---------------------------|----------|-----------------|------------|---------|
| Temp C                    | Tol Flow | Tol Composition | Tol Amount | Unit    |
| 130                       | 10.74    | 0.3433          | 3.687042   | kmol/hr |
| 132                       | 41.29    | 0.3438          | 14.195502  | kmol/hr |
| 134                       | 68.97    | 0.3478          | 23.987766  | kmol/hr |
| 136                       | 96.44    | 0.5799          | 55.925556  | kmol/hr |
| 138                       | 131.8    | 0.4313          | 56.84534   | kmol/hr |
| 140                       | 183.7    | 0.5253          | 96.49761   | kmol/hr |
| 142                       | 240.1    | 0.5784          | 138.87384  | kmol/hr |
| 144                       | 309.8    | 0.5677          | 175.87346  | kmol/hr |

| Varying R from 1.5 to 2.5   T =142 C |          |          |            |         |
|--------------------------------------|----------|----------|------------|---------|
| R                                    | Tol Flow | Tol Comp | Amount Tol | Unit    |
| 1.5                                  | 280.8    | 0.5093   | 143.01144  | kmol/hr |
| 1.6                                  | 274.8    | 0.5181   | 142.37388  | kmol/hr |
| 1.7                                  | 269.5    | 0.5264   | 141.8648   | kmol/hr |
| 1.8                                  | 264.6    | 0.5344   | 141.40224  | kmol/hr |
| 1.9                                  | 260.1    | 0.5419   | 140.94819  | kmol/hr |
| 2                                    | 256      | 0.549    | 140.544    | kmol/hr |
| 2.1                                  | 252.3    | 0.5557   | 140.20311  | kmol/hr |
| 2.2                                  | 248.8    | 0.562    | 139.8256   | kmol/hr |
| 2.3                                  | 245.7    | 0.5679   | 139.53303  | kmol/hr |
| 2.4                                  | 242.9    | 0.5734   | 139.27886  | kmol/hr |
| 2.5                                  | 240.1    | 0.5784   | 138.87384  | kmol/hr |